James Cayson

College of Information Science

IS 523

Professor: John Chen

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Andromeda Final Project Write Up

Intro

I decided to do my final project on predicting how many planets might be in the Andromeda Galaxy using a Random Forest and Multi Layer Perceptron. Since there are no confirmed planets in Andromeda we must use what we understand about the Milky Way. For this reason, I decided to use data from the NASA exoplanet archive and general understandings of cosmology to extrapolate that data to what we know about Andromeda. This namely being the stars in Andromeda as we do know some information about that since they are so big and luminous. Therefore, I downloaded the data from NASA and cleaned for variables "hostname", "sy_snum", "sy_pnum", "st_teff", "st_mass", "st_rad", "st_met". Hostname is for star, sy_num is for number of planets in the planetary system, sy_pnum is for number of confirmed planets in the system, st_teff is the stars's temperature in Kelvin, st_mass is the star mass, st_rad is the radius of the star, and st_met is the metal in the star. I needed to clean this data because planet formation around stars is largely directly or indirectly determined by these variables. I also had to clean these variables themselves or risk inaccuracy in my models like for instance with sy_snum which would have caused duplicates in the data due to planets orbiting multiple stars. For the Andromeda star data, I originally wanted to use real PHAT data from Andromeda, but it was outside the scope of this assignment unfortunately. This was due to the file size being too big for Github and requiring a graduate level astrophysics education to decode data to match the Milky Way. Consequently, I made a synthetic data set that took into account well known average data of Andromeda's stars. This enabled my Milky Way data to be engineered and scaled to match the synthetic dataset.

Justification for Approach

I chose to use Random Forest and Multi-Layer Perceptron here. I chose a random forest because it sets a solid baseline for how a data pipeline should perform. It enables a user to see the robust nonlinear relationships and allows for minimal assumptions in regards to data distribution. A MLP was also implemented to model more complex feature interactions. The MLP models the relationship between stellar properties and planet counts by learning a non-linear continuous function through multiple hidden layers trained via backpropagation. Also, using multiple models allows for performance comparison and increases confidence in the results.

Code

I wanted the code to be very clean and easy to understand. This is why I organized the code into clear sections covering data loading, data cleaning, model training, scaling, and evaluation. Descriptive variable names and consistent formatting were used to improve readability. Comments and section headers document key steps, particularly where data transformations and modeling decisions were also applied. One such example of this is cell 7 where I specifically included extra lines of code to ensure that those unfamiliar with star types by temp in Kelvin could easily pick up which stars they are. Another area that I felt like I took into consideration the readability of my code included cell 17 (synthetic dataset) and cell 23-25 (results). Beyond overall organization, serious care was taken to ensure that each section of the notebook could be understood independently by someone reviewing the work for the first time. I also made sure that data transformations were applied incrementally other than in a single condensed step to make it easier for people to trace how raw values were converted into modeling-ready features.

Mining Method

My Random Forest was trained on 300 decision trees which were applied to the unscaled data set. The MLP was trained on standardized features in the Milky Way data set to ensure stable and efficient learning. The dataset was split into training and testing subsets using the same split for both models to allow for a fair comparison. In addition, using the same training and testing split for both models ensured that performance differences were attributable to the modeling approach rather than variations in data exposure. I believe this methodological consistency strengthens the validity of the comparison. Through storing predictions separately for each model, the analysis allows for direct comparison of outputs when applied to the synthetic Andromeda dataset. This enables for a deeper understanding of the differences between the mining methods.

Discussion

The Random Forest and Multi-Layer Perceptron models both had good predictive performance. The Random Forest achieved a high R^2 score of .9890113005110388 and a low MSE of .00781545213291733. The MLP had a R^2 score of .9719094427088885 and a MSE of .01997874326404107. This indicates that the selected stellar features explain a good amount of variance in estimated planet counts. Furthermore, it suggests that stellar temperature, mass, radius, and metallicity are meaningful predictors for modeling planet formation trends. In addition, the low MSE indicates that the models' predictions are only slightly different from the observed values on average, suggesting high numerical accuracy rather than just strong relative fit.

The Random Forest produced slightly lower but more consistent estimates compared to the MLP, predicting an average of approximately 3.32 planets per star. When scaled to the Andromeda galaxy, this corresponds to an estimated total of about 3.32 trillion planets, including roughly 1.33 trillion rocky planets and 332 billion habitable-zone planets. The MLP produced slightly higher estimates compared to the Random Forest, predicting an average of approximately 3.35 planets per star. When scaled to the Andromeda galaxy, this corresponds to an estimated total of about 3.35 trillion planets, including roughly 1.34 trillion rocky planets and 335 billion habitable-zone planets. Overall, I think both models increased confidence in the predicted results when applied to the Andromeda synthetic dataset, while still acknowledging that the estimates represent theoretical predictions rather than direct observations. This, again, is why a Random Forest and a Multi-Layer Perceptron were used together to ensure validity. Finally, thank you for reviewing my assignment.